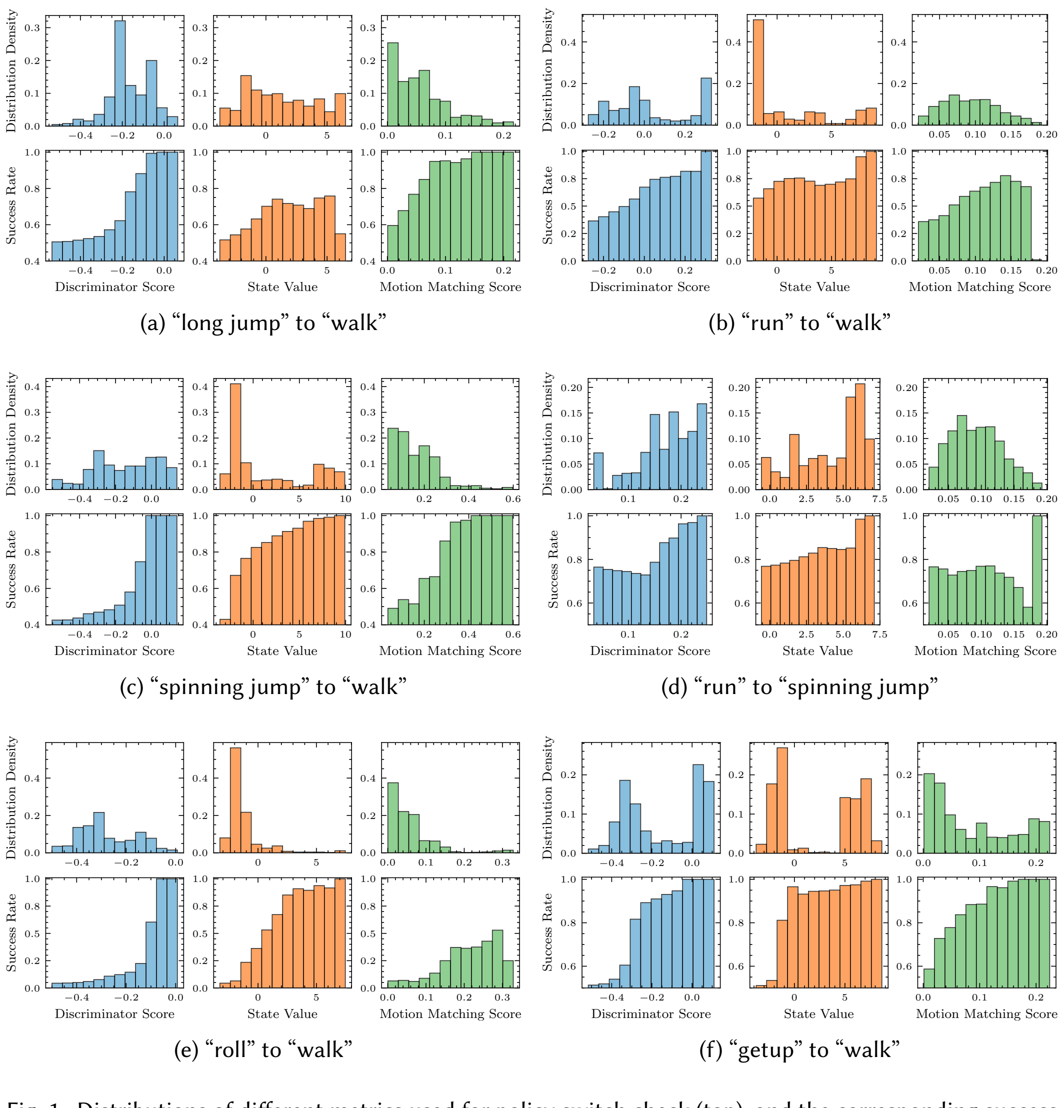
A GAN-Like Approach for Physics-Based Imitation Learning and Interactive Control – Supplementary material

1 REFERENCE MOTION CLIP ACQUISITION

Motion Clip	Collection	File	Start	End
walk		run2_subject1	5758	5791
pace		walk1_subject1	3748	3799
limp		walk1_subject1	4515	4572
swaggering walk		walk2_subject1	2821	2853
sashay walk		walk2_subject1	2613	2645
jaunty walk		walk2_subject1	5002	5044
stomp walk		walk2_subject1	6581	6618
stoop walk		walk4_subject1	986	1014
joyful walk		dance1_subject1	2018	2054
walk with arms akimbo		walk2_subject1	6454	6520
sharp turn during running		run2_subject1	435	479
90-degree turn during walking		walk1_subject1	3386	3448
roll		pushAndStumble1_subject2	4679	4780
fight stance		fightAndSports1_subject4	1096	1110
run	run	run2_subject1	5779	5850
run 2		run2_subject1	5537	5608
spinning jump	spinning and long jump	jumps1_subject1	6682	6772
spinning jump 2		jumps1_subject1	6858	6949
long jump		multipleActions1_subject1	3234	3287
long jump 2		multipleActions1_subject1	3748	3800
get up	get up	fallAndGetUp3_subject1	1318	1404
get up 2		fallAndGetUp3_subject1	2016	2100
get up 3		fallAndGetUp3_subject1	2247	2326
get up 4		fallAndGetUp3_subject1	2473	2580
get up 5		fallAndGetUp3_subject1	2717	2772
punch	punch	fightAndSports1_subject4	1243	1295
punch 2		fightAndSports1_subject4	1109	1148
punch 3		fightAndSports1_subject4	1317	1369
kick	kick	fightAndSports1_subject4	1394	1464
kick 2		fightAndSports1_subject4	971	1036
kick 3		fightAndSports1_subject4	1037	1099
kick 4		fightAndSports1_subject4	1493	1549

Table 1. Reference motion clips acquired from the LAFAN1 dataset. `Start` and end denote the start and end frame, respectively, of the corresponding file in the LAFAN1 dataset from which a reference clip is obtained.

2 POLICY SWITCH SUCCESS RATE VERSUS DISCRIMINATOR SCORE



(a) "long jump" to "walk"

(b) "run" to "walk"

(c) "spinning jump" to "walk"

(d) "run" to "spinning jump"

(e) "roll" to "walk"

(f) "getup" to "walk"

Fig. 1. Distributions of different metrics used for policy switch check (top), and the corresponding success rate with different values used as the switch threshold value (bottom). See Section 6.5 for details of the experiment.